



Fall 2026 Semester
Cryptographic Application

Setting for Windows

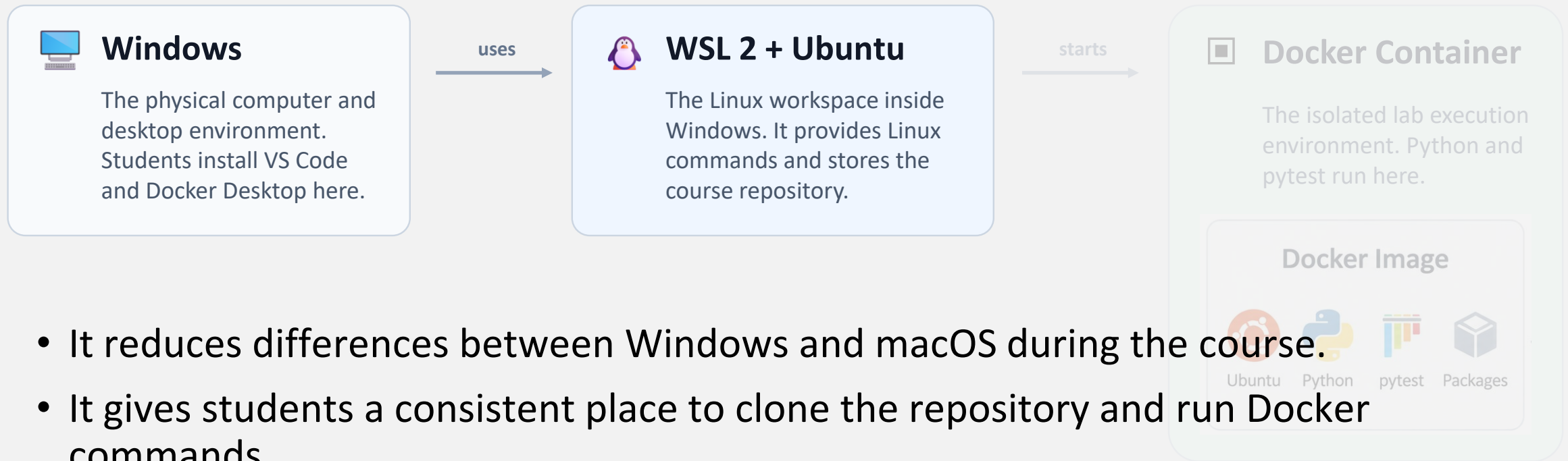
LIBERTAS
JUSTITIA
VERITAS

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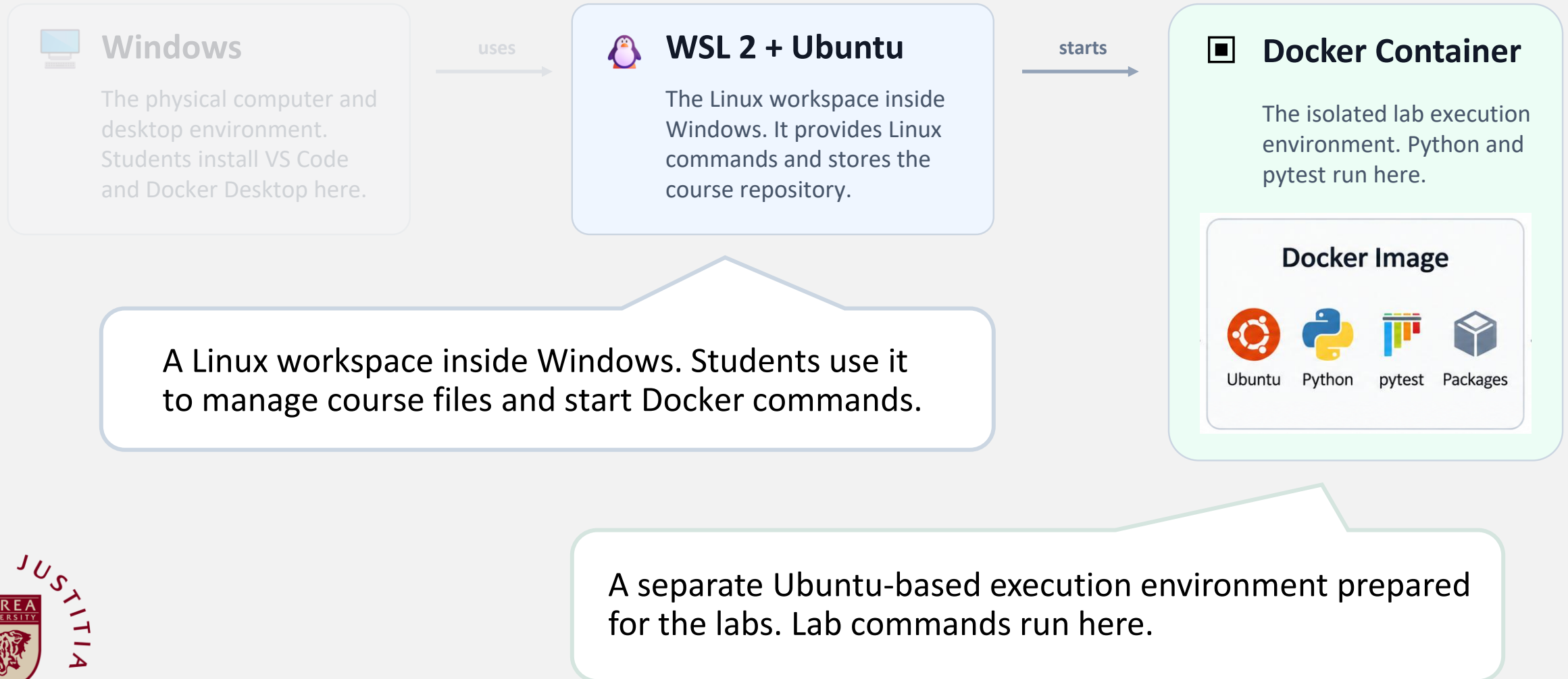
The Windows lab stack



Why do Windows students install WSL?

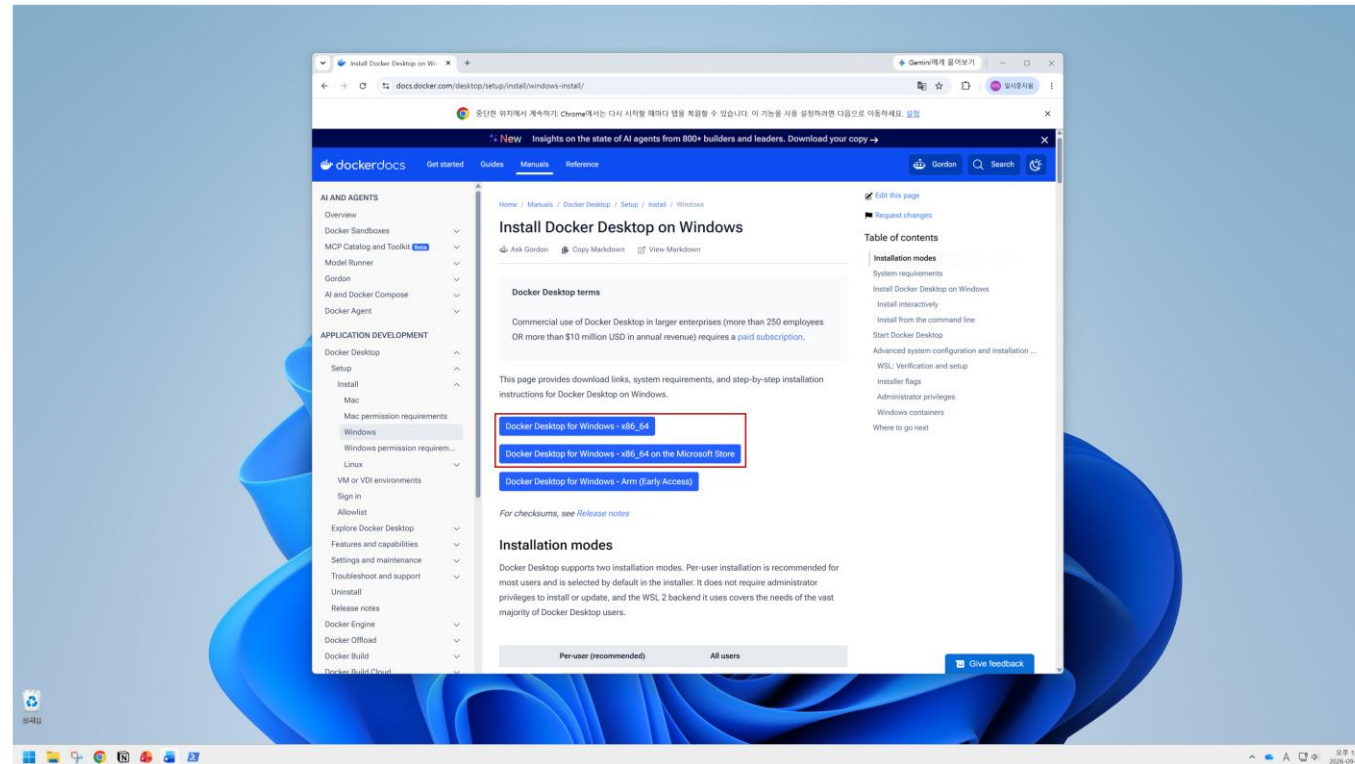


WSL Ubuntu vs. Docker Ubuntu



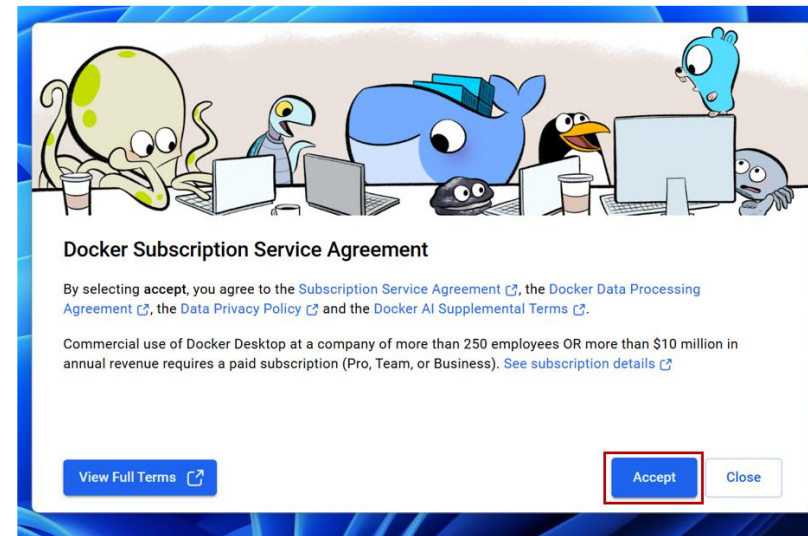
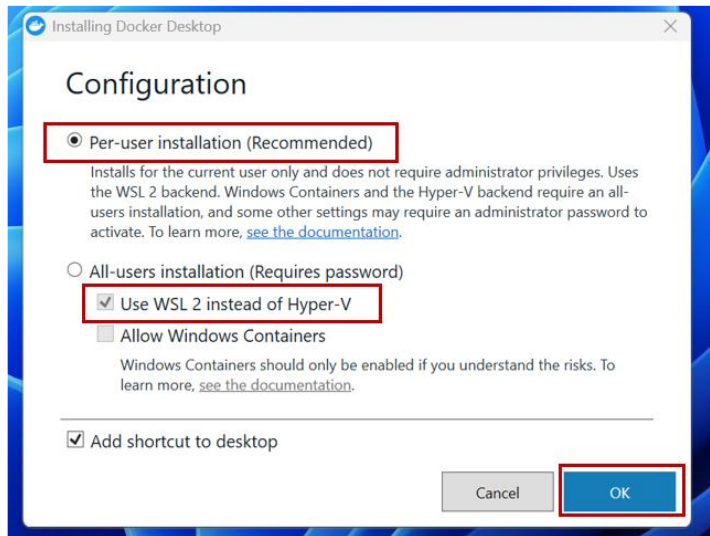
1. Install Docker Desktop

- Open the official Docker Desktop installation page for Windows.
→ <https://docs.docker.com/desktop/setup/install/windows-install/>
- Download the Docker Desktop installer.



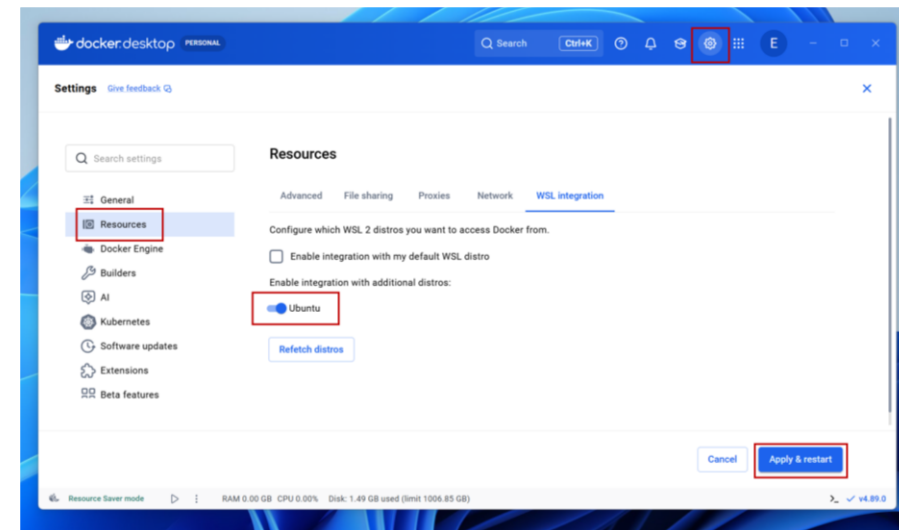
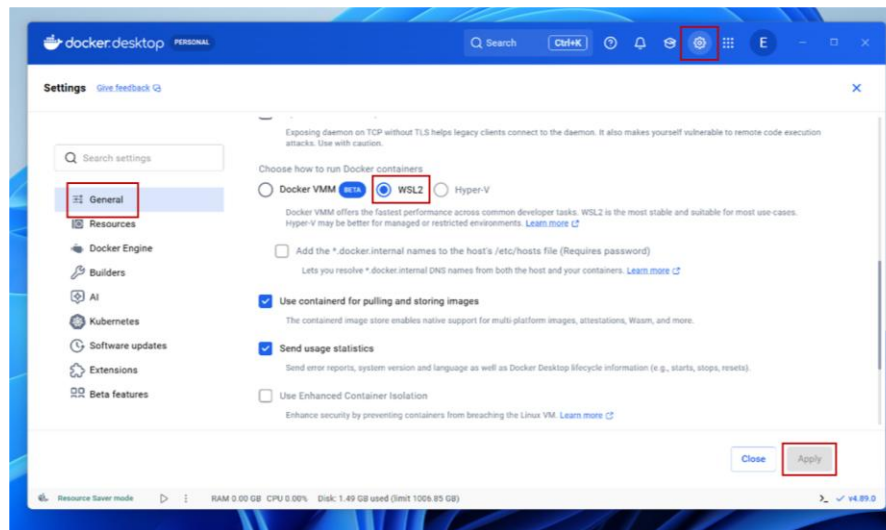
2. Complete the Docker Desktop Installation

- Run the installer.
- If the installer shows an option to use WSL 2, make sure it is selected.
- Otherwise, continue with the default settings.



3. Configure Docker Desktop

- Open Settings > General and select WSL 2 under Choose how to run Docker containers.
- Open Settings > Resources > WSL Integration, then enable the Ubuntu distribution that you will use for the course.



4. Install WSL 2

- Open **PowerShell as Administrator**. Run:

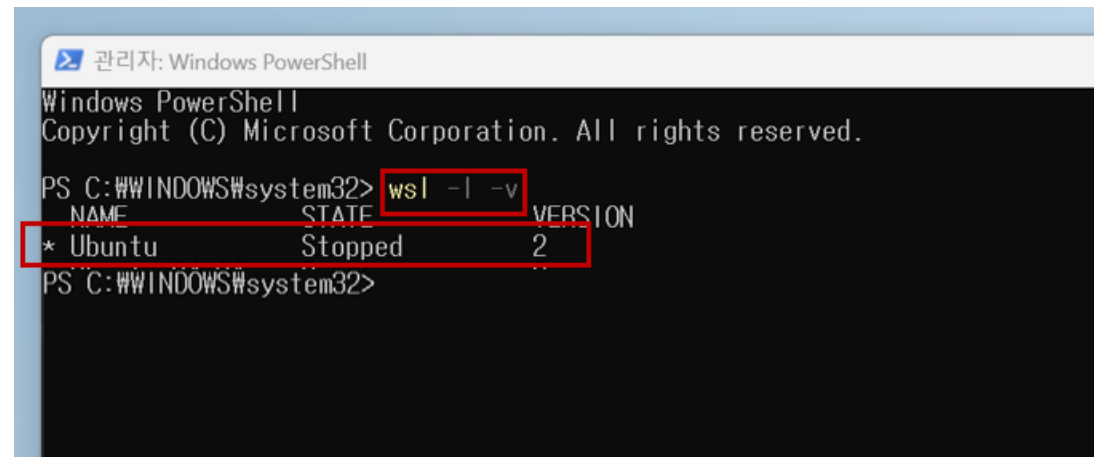
```
wsl --install
```

- Check whether Ubuntu is installed:

```
wsl -l -v
```

If **Ubuntu** does not appear in the list, then install Ubuntu:

```
wsl --install -d Ubuntu
```

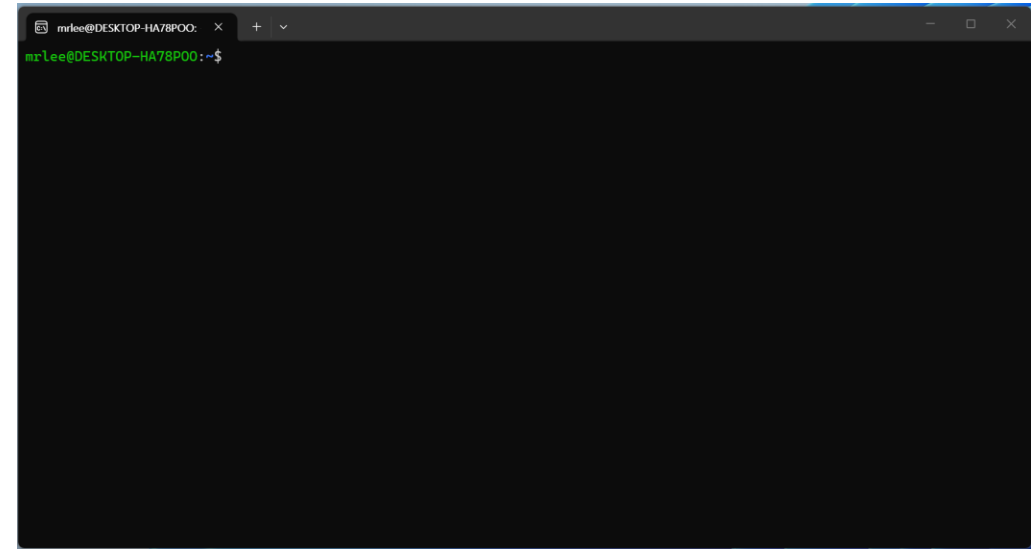
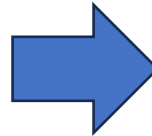
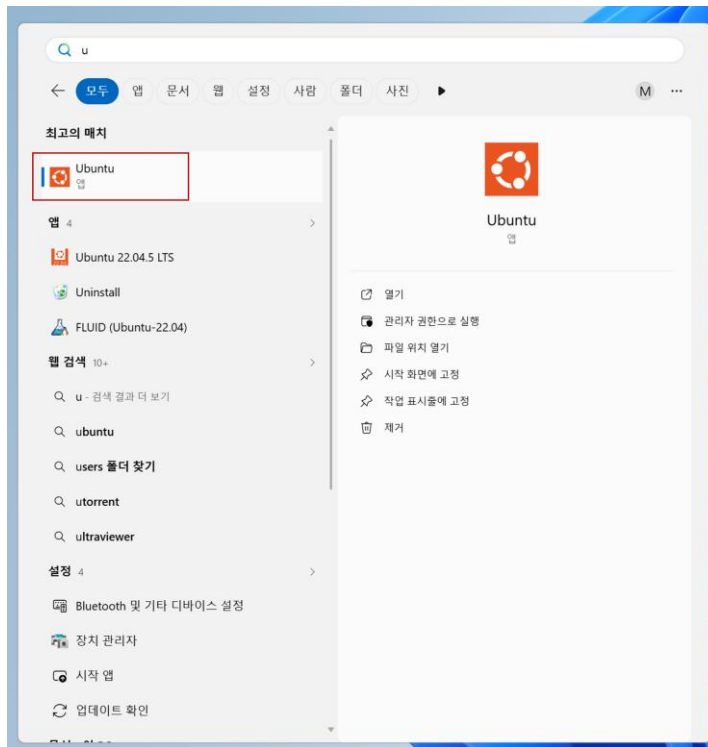


A screenshot of a Windows PowerShell terminal window titled "관리자: Windows PowerShell". The terminal shows the command `wsl -l -v` being executed, which lists installed Windows Subsystem for Linux (WSL) distributions. The output is a table with three columns: NAME, STATE, and VERSION. The table shows one entry: * Ubuntu, Stopped, 2. The command prompt is `PS C:\WINDOWS\system32>`.

NAME	STATE	VERSION
* Ubuntu	Stopped	2

5. Open the Ubuntu Terminal

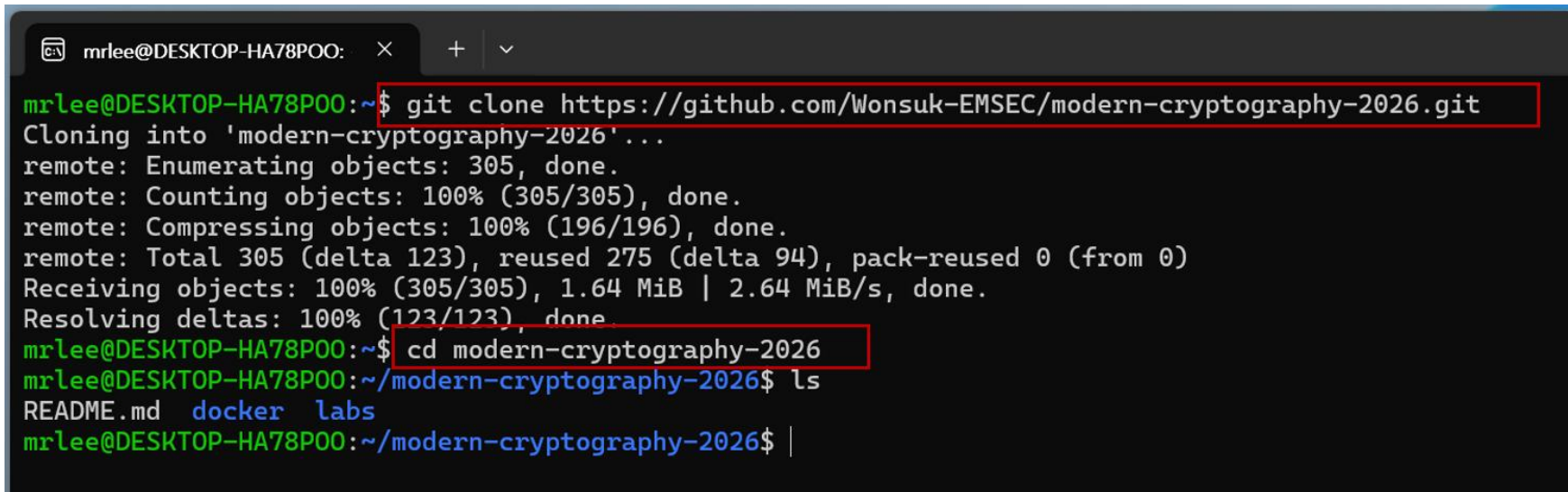
- Launch the Ubuntu terminal and complete the initial user setup if prompted.



6. Clone the Course Repository

- Run in: HOST - Ubuntu Terminal

```
cd ~  
git clone https://github.com/Wonsuk-EMSEC/modern-cryptography-2026.git  
cd modern-cryptography-2026
```



A screenshot of a terminal window with a dark background. The window title is 'mrlee@DESKTOP-HA78POO:'. The terminal shows the following commands and output:

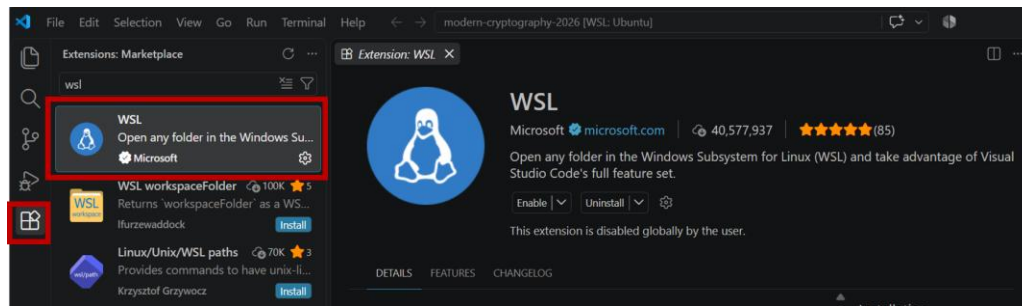
```
mrlee@DESKTOP-HA78POO:~$ git clone https://github.com/Wonsuk-EMSEC/modern-cryptography-2026.git  
Cloning into 'modern-cryptography-2026'...  
remote: Enumerating objects: 305, done.  
remote: Counting objects: 100% (305/305), done.  
remote: Compressing objects: 100% (196/196), done.  
remote: Total 305 (delta 123), reused 275 (delta 94), pack-reused 0 (from 0)  
Receiving objects: 100% (305/305), 1.64 MiB | 2.64 MiB/s, done.  
Resolving deltas: 100% (123/123), done.  
mrlee@DESKTOP-HA78POO:~$ cd modern-cryptography-2026  
mrlee@DESKTOP-HA78POO:~/modern-cryptography-2026$ ls  
README.md  docker  labs  
mrlee@DESKTOP-HA78POO:~/modern-cryptography-2026$ |
```

Red boxes highlight the clone command and the directory change command.

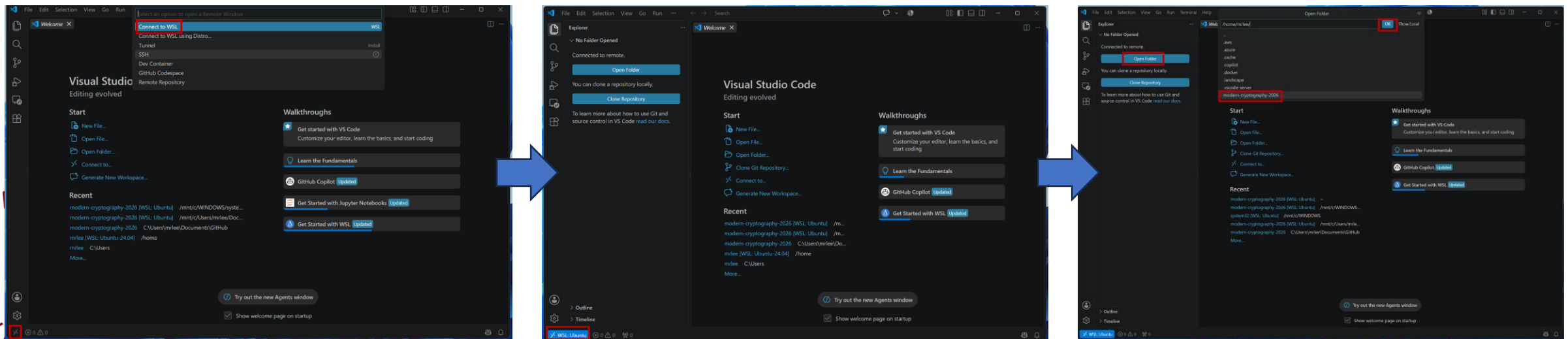


7. Open the Course Repository in VS Code

- Install the WSL extension in Visual Studio Code if it is not already installed.

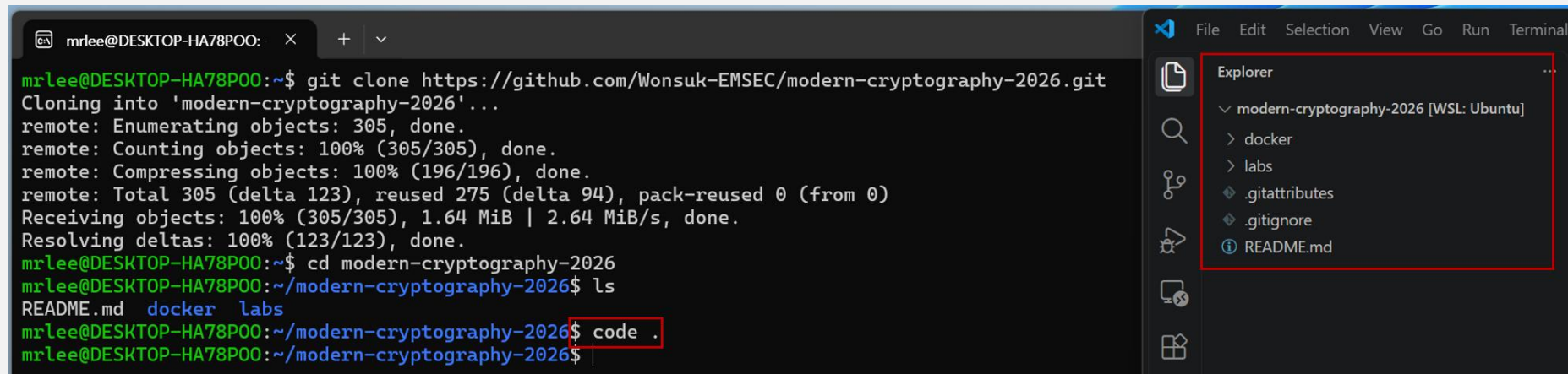


- Open the course repository in VS Code.



7. Open the Course Repository in VS Code

- From the Ubuntu terminal, run: `code .`



The screenshot shows the Visual Studio Code (VS Code) interface. On the left, a terminal window displays the following commands and output:

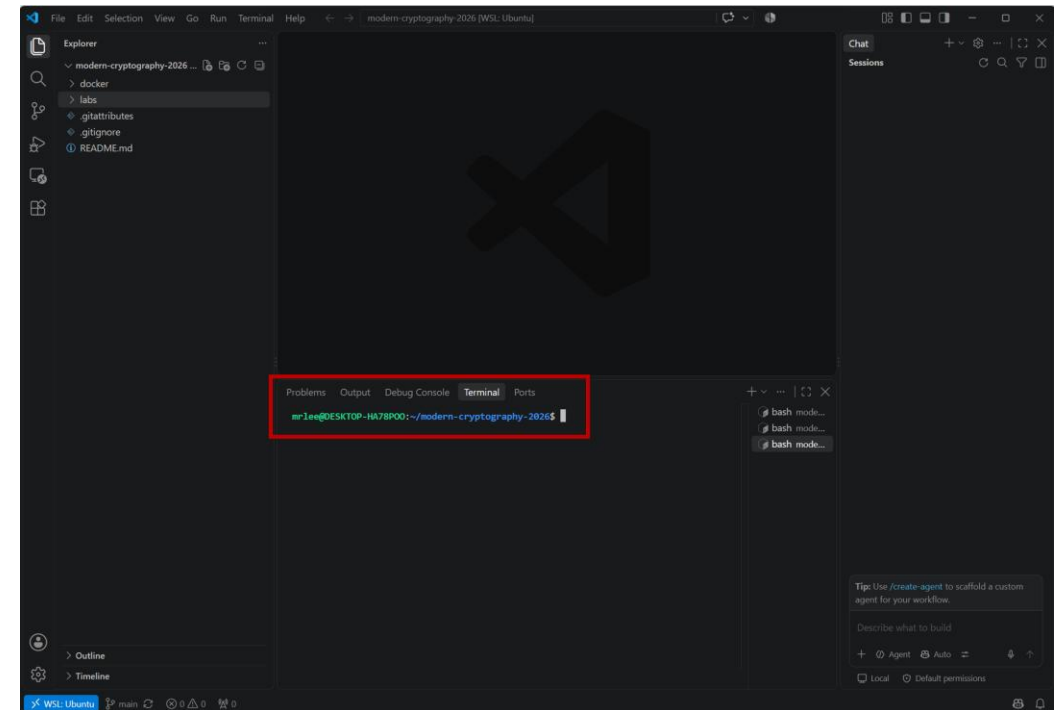
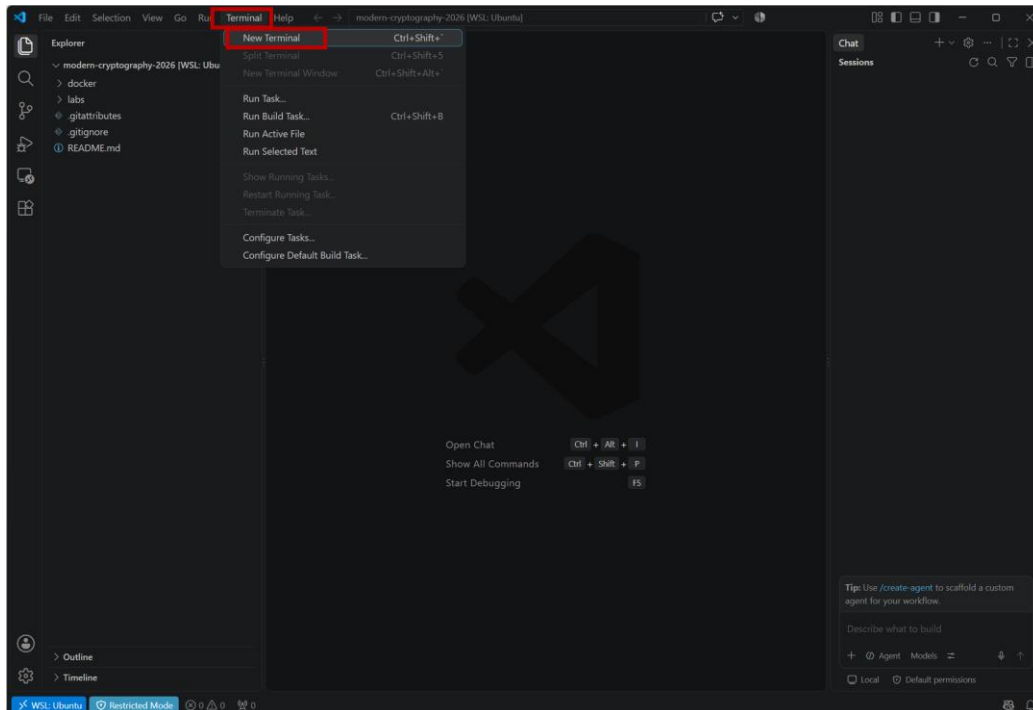
```
mrlee@DESKTOP-HA78POO:~$ git clone https://github.com/Wonsuk-EMSEC/modern-cryptography-2026.git
Cloning into 'modern-cryptography-2026'...
remote: Enumerating objects: 305, done.
remote: Counting objects: 100% (305/305), done.
remote: Compressing objects: 100% (196/196), done.
remote: Total 305 (delta 123), reused 275 (delta 94), pack-reused 0 (from 0)
Receiving objects: 100% (305/305), 1.64 MiB | 2.64 MiB/s, done.
Resolving deltas: 100% (123/123), done.
mrlee@DESKTOP-HA78POO:~$ cd modern-cryptography-2026
mrlee@DESKTOP-HA78POO:~/modern-cryptography-2026$ ls
README.md  docker  labs
mrlee@DESKTOP-HA78POO:~/modern-cryptography-2026$ code .
mrlee@DESKTOP-HA78POO:~/modern-cryptography-2026$
```

The Explorer sidebar on the right shows the file structure of the cloned repository, with a red box highlighting the contents:

- modern-cryptography-2026 [WSL: Ubuntu]
 - > docker
 - > labs
 - ◆ .gitattributes
 - ◆ .gitignore
 - ◆ README.md



8. Open the Visual Studio Code Terminal



Check that “WSL: Ubuntu” appears in the bottom-left corner.



9. Build and Start the Course Container

- The first time you use the course environment, build the Docker image:

```
docker compose -f docker/compose.yml build
```

→ Docker Desktop must be running before you use Docker commands.

- After the image has been built, start the course container:

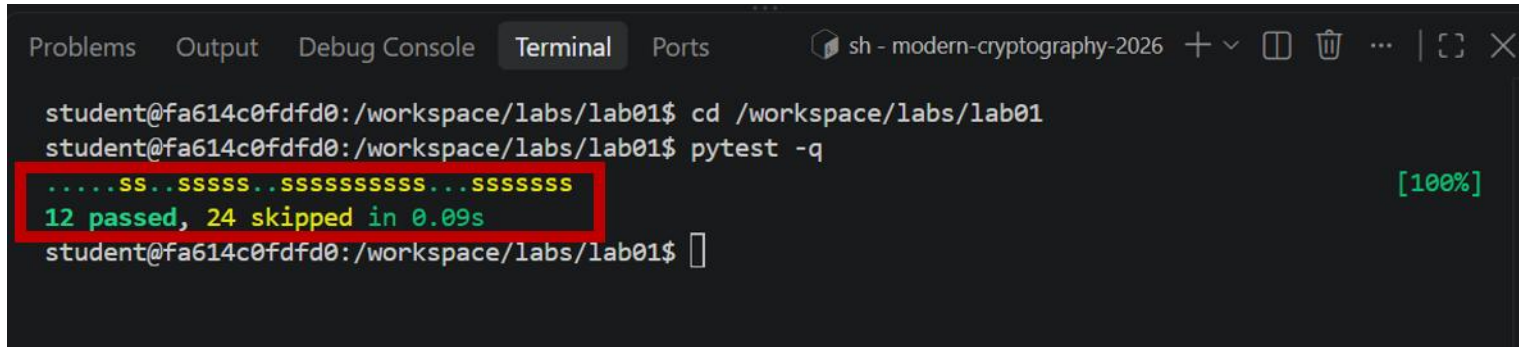
```
docker compose -f docker/compose.yml run --rm course bash
```



10. Test the Lab Environment

- Run:

```
cd /workspace/labs/lab01  
pytest -q
```



The screenshot shows a terminal window with the following content:

```
Problems  Output  Debug Console  Terminal  Ports  sh - modern-cryptography-2026 + v [ ] [ ] ... | [ ] [ ] X  
  
student@fa614c0dfd0:/workspace/labs/lab01$ cd /workspace/labs/lab01  
student@fa614c0dfd0:/workspace/labs/lab01$ pytest -q  
.....ss..sssss..ssssssssss...sssssss [100%]  
12 passed, 24 skipped in 0.09s  
student@fa614c0dfd0:/workspace/labs/lab01$
```

The output line `.....ss..sssss..ssssssssss...sssssss` is highlighted with a red box. The summary line `12 passed, 24 skipped in 0.09s` is also highlighted with a red box. The progress indicator `[100%]` is shown in green.

